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Causal Disentanglement Domain Generalization for time-series signal fault diagnosis

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ABSTRACT

Domain generalization-based fault diagnosis (DGFD) presents significant prospects for recognizing faults without the accessibility of the target domain. Previous DGFD methods have achieved significant progress; however, there are some limitations. First, most DGFG methods statistically model the dependence between time-series data and labels, and they are superficial descriptions to the actual data-generating process. Second, most of the existing DGFD methods are only verified on vibrational time-series datasets, which is insufficient to show the potential of domain generalization in the fault diagnosis area. In response to the above issues, this paper first proposes a DGFD method named Causal Disentanglement Domain Generalization (CDDG), which can reestablish the data-generating process by disentangling time-series data into the causal factors (fault-related representation) and non-causal factors (domain-related representation) with a structural causal model. Specifically, in CDDG, causal aggregation loss is designed to separate the unobservable causal and non-causal factors. Meanwhile, the reconstruction loss is proposed to ensure the information completeness of the disentangled factors. We also introduce a redundancy reduction loss to learn efficient features. The proposed CDDG is verified on five cross-machine vibrational fault diagnosis cases and three cross-environment acoustical anomaly detection cases by comparing it with eight state-of-the-art (SOTA) DGFD methods. We publicize the open-source time-series DGFD Benchmark containing CDDG and the eight SOTA methods. The code repository will be available at <https://github.com/ShaneSpace/DGFDBenchmark>.

1. Introduction

Fault diagnosis is increasingly important in ensuring safe operation of machines and reducing unexpected breakdown in critical equipment (Su et al., 2022). Traditional fault diagnosis methods are based on signal processing or machine learning algorithms. These methods rely on the knowledge and experiments of experts or need to design features of time-series data manually (Jia, Chow, Yu, & Yuan, 2022). Deep learning-based fault diagnosis has become popular due to its powerful ability to extract discriminative features automatically. Various neural network models, such as convolutional neural networks (CNNs) (Jia, Chow, & Yuan, 2023), graph convolutional networks (Yu, Tang, & Zhang, 2021), and deep belief networks (Zhao et al., 2022), were developed for diagnosing machine faults. Most classical deep learning-based methods assume that the real operations follow the same distribution as the training dataset. However, when there is a distribution discrepancy, these models may suffer severe performance degradation. Time-series data distribution can vary easily because of many factors, such as working conditions, sensor placements, and background noises. As a

result, domain shift problem (Li, Zhang, Ma, Luo, & Li, 2020b) occurs, causing the trained models fail in recognizing faults.

The inept of deep learning-based methods under domain shift induced the research of domain adaptation fault diagnosis (DAFD) (Tian, Han, Karimi, Zhang, & Shi, 2023). DAFD assumes that the labeled source domain (training) data and unlabeled target domain (test) data are available in the training stage. The goal of DAFD is to realize cross-domain fault diagnosis on the target domain by aligning the distribution of target domain data with given source domain data. For example, Liu, Wang, Chow, and Li (2022) proposed a subdomain adaptation method that could align subdomain distributions by minimizing the local maximum mean discrepancy and global distributions by adversarial training. Shi et al. (2022) presented a multi-source domain factorization network for cross-working-condition and cross-component fault diagnosis. In Jiao, Lin, Zhao, Liang, and Ding (2022), a cycle-consistent adversarial adaptation network was developed for cross-domain fault diagnosis. In this method, adversarial strategy was used to induce transferable feature learning. Although domain shift can

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be mitigated to some extent, the DAFD methods need the accessibility of unlabeled target domain data, which can be difficult and expensive to collect. Additionally, the trained DAFD model can only be suitable for the specific target domain [Chen et al. \(2021\)](#), and they need to be retrained when applied to a new target domain.

Motivated by the above drawbacks, domain generalization-based fault diagnosis (DGFD) ([Zhao & Shen, 2023b](#)) was introduced because it does not require target domain data for model training. Once it is trained, the model can be applied to any potential target domains. In DGFD, it is assumed that different domains share certain “stable” or “domain-invariant” features, and the goal of DGFD is to learn those features and to generalize the trained model to unseen target domains. The core objective of domain generalization is to learn domain-invariant features ([Zhang, Zhao, et al., 2021](#)). To achieve this objective, many DGFD methods were proposed to model the statistical dependence between data and labels by learning representations that are marginally or conditionally independent of domains. Bearings are critical components of rotating machines, and their vibration signals are challenging time-series signal to handle. Bearing fault diagnosis is indeed a crucial application area in machine condition monitoring. Bearing domain generalization-based fault diagnosis is of particular appealing when dealing with the significant domain shift caused by varying working conditions and machine structures. Faulty bearing time-series signals, which contain periodic impulses generated by different kinds of fault of different bearings, exhibit various distributions making fault diagnosis difficult. For example, [Yang et al. \(2020\)](#) developed a DGFD method called CNN-C which could learn generalization features with a center loss and identify faults of machines working in unseen operating conditions. [Zhao and Shen \(2022\)](#) proposed a domain generalization network combining invariance and specificity (DGNIS), in which triplet loss was used to align the class-conditional distribution. In their model ([Zhao & Shen, 2022](#)), CORAL loss was adopted to reduce the marginal distribution gap between signals of different working conditions. In [Ragab et al. \(2022\)](#), a conditional contrastive domain generalization (CCDG) was devised for rotating machinery fault diagnosis. The method could learn domain-independent representation by maximizing the mutual information of similar classes across different domains while minimizing the mutual information among different classes. In [Zheng et al. \(2020\)](#), a deep domain generalization network for fault diagnosis (DDGFD) was proposed. The method could realize cross-machine fault diagnosis by combining a priori diagnosis knowledge, and domain-invariant features were extracted via the minimization of instance-based discriminative loss. [Han, Li, and Qian \(2021\)](#) reported an intrinsic and extrinsic domain generalization network (IEDGNet). The method facilitated intrinsic generalization by minimizing triplet loss and enhances extrinsic generalization with an adversarial training strategy. In [Li, Zhang, Ma, Luo, and Li \(2020a\)](#), the domain-invariant representation was extracted by combining adversarial training with data augmentation based on signal stretching.

Despite the promising results achieved by the above statistical dependence-based methods, it may still be insufficient to provide strong generalization to unseen distributions because of the lack of data-generating mechanism ([Lv et al., 2022](#)). Meanwhile, models may still be biased by the spurious correlation in data ([Miao, Yuan, & Kuang, 2022](#)). In light of the causal invariance mechanism, causality-inspired DGFD methods have garnered growing interest. For example, [Li, Wang, Zi, and Zhang \(2021\)](#) presented a Whitening-Net for cross-machine bearing fault diagnosis. In this method, a causal loss was designed to impose regularization constraints on the network so that the causality between data-label pairs from multiple conditions can be extracted. In [Li, Wang, Zi, Zhang, and Li \(2022\)](#), a causal consistency network (CCN) was proposed for bearing fault diagnosis. The method could divulge invariant causal information through the consistency regularizations of feature extractor and causal predictor. Both Whitening-Net and CCN are grounded in causal learning. Nevertheless, they solely take causal factors into account while neglecting non-causal factors.

Thus, the spurious correlation may not be removed sufficiently. More recently, [Jia, Li, Wang, Sun, and Deng \(2023\)](#) proposed a deep causal factorization network (DCFN) for diagnosing faults of different machines. In the DCFN, the causal task factorization module and the feature factorization module are devised to separate causal factors (fault-related representation) and non-causal factors (domain-related representation). However, the DCFN overlooks the information completeness of causal and non-causal factors, and it may result in trivial solutions. Besides, it is worth noting that all the causal-based DGFD methods mentioned above have only been verified on vibration-based faulty signal diagnosis cases. Therefore, their scope of applicability is limited. In addition, there is no open-source DGFD benchmark in the fault diagnose field. Thus, it is difficult for researchers to conduct the experimental verification and thorough comparative analysis.

To tackle the above challenges, this paper proposes a new DGFD method called Causal Disentanglement Domain Generalization (CDDG), in which the data-generating process is rebuilt, and the causal and non-causal factors are disentangled according to the structural causal model (SCM). Concretely, causal aggregation loss is devised to decouple causal factors from non-causal factors by enforcing data from the same classes to aggregate together. Meanwhile, we make causal and non-causal factors work well and avoid trivial solutions by optimizing the reconstruction loss. This is significant because it can solve information loss problem in the disentanglement process. Also, inspired by the redundancy-reduction principle in Barlow Twins ([Zbontar, Jing, Misra, LeCun, & Deny, 2021](#)), the redundancy reduction loss is introduced to learn efficient features. Rather than relying solely on experimental validation on vibrational time-series bearing faults diagnosis across different machines, we also conduct a comprehensive set of experiments that include acoustical fan anomaly detection. Our results provide comparative analysis against the state-of-the-art (SOTA) methods. Finally, it is noted that we release a code repository that includes the proposed CDDG and eight SOTA methods all integrated into a unified framework. To the best of our knowledge, this is the first open-source benchmark in the DGFD research. The major contributions of the paper are summarized as follows:

- (1) This paper proposes a new DGFD method called CDDG, which attempts to learn invariant diagnostic knowledge by rebuilding the data-generating process based on SCM. Causal and non-causal factors can be disentangled by optimizing causal aggregation loss, reconstruction loss, and redundancy reduction loss simultaneously. Specifically, the causal aggregation loss enables the model to separate causal and non-causal factors. The reconstruction loss ensures information completeness and avoid trivial solutions. The redundancy reduction loss suppresses the correlation between the feature dimensions of factor vectors and decreases entanglement between causal and non-causal factors.

- (2) This paper conducts extensive experiments to verify the effectiveness of CDDG. Compared with eight SOTA DGFD methods, the proposed CDDG outperformed them on cross-machine vibrational bearing fault diagnosis and cross-environment acoustical fan anomaly detection. These two tasks are typical domain generalization applications in real-world engineering. The domain shift owing to machine characteristics and environmental changes in these two tasks are highly challenging and representative. Our proposed CDDG is able to provide a strong generalization ability to solve challenging cross-machine and cross-environment time-series fault diagnosis problems.

- (3) This paper releases the first open-source code library of the DGFD benchmark. The comprehensive library is built on a unified framework and is convenient to extend. It will facilitate domain generalization-based fault diagnosis research.

The rest of the paper is organized as follows. In Section 2, the preliminaries of causal learning are given. The proposed CDDG is introduced in Section 3. In Section 4, the proposed method is verified on cross-machine bearing fault diagnosis tasks and cross-environment anomaly detection tasks. In Section 5, we give the limitations of the method and future works. Finally, Section 6 presents the conclusion.

2. Preliminaries

2.1. Causal learning for domain generalization

Causality is a finer description of variable relations beyond statistics (Wang, Lan, et al., 2022). Generally, causal learning focuses on the fact that the traditional conditional probabilities in statistics cannot give reliable predictions under intervention (Schölkopf, Locatello, Bauer, Ke, Kalchbrenner, Goyal, & Bengio, 2021). Interestingly, the domain shift problem in domain generalization (DG) can also be regarded as the intervention that brings a spurious correlation between irrelevant features and category labels (Zhang, Cui, et al., 2021). Thus, causal learning was introduced to solve the DG problem. Many causality-based DG methods were proposed. For example, Lv et al. (2022) proposed a causality-inspired representation learning (CIRL) method, which conducted intervention by disturbing the amplitude of Frontier transformation and learned causal features via three basic properties. Mahajan, Tople, and Sharma (2021) developed a match-based DG (MatchDG) algorithm that constructed a structural causal model and enforced the multi-domain samples from the same objects to map to the same representations. The causality-based DG methods are based on the causal invariance principle, which indicates that the data from multiple domains share the same generation mechanism across domains (Liu et al., 2021). Therefore, it is promising to use causal learning for the domain generalization especially on challenging fault diagnosis problem.

2.2. Signal generation mechanism and structural causal model for fault diagnosis

In fault diagnosis, the collected signal $x(t)$ can be modeled as the summation of the label-specific causal factor $\delta(t) * h^\delta(t)$ and the domain-specific non-causal factor $d(t)h^d(t)$. That is,

$$x(t) = \delta(t) * h^\delta(t) + d(t)h^d(t), \quad (1)$$

where $*$ denotes convolution operation, $\delta(t)$ is the impulse due to machine fault, $d(t)$ is the disturbance owing to the domain shift, including the background noise, the sensor type, the sampling frequency of the signal, the working condition (i.e., the rotating speed and the load), and the machine structure, etc. $h^\delta(t)$ and $h^d(t)$ are the impulse response functions for $\delta(t)$ and $d(t)$, respectively. It is assumed that the item $\delta(t) * h^\delta(t)$ does not contain any information on domain-related information, and the item $d(t)h^d(t)$ does not contain any fault-related information. This assumption indicates that fault-related causal factor $\delta(t) * h^\delta(t)$ and the domain-related noncausal factor $d(t)h^d(t)$ can be disentangled.

The causal relationship in above signal generation mechanism can be described by the structural causal model (SCM), which is a directed acyclic graph (DAG). In an SCM, each node represents a variable, and each edge is a function. If node A is the child of node B , we can say A is caused by B . In fault diagnosis, the SCM describing the data-generating process has been well studied by Jia, Li, et al. (2023) and Li, Wang, Zi, Zhang, and Wan (2021), as shown in Fig. 1. Suppose that ϵ_X , ϵ_Y , ϵ_{Y_D} , ϵ_{Z_C} , and ϵ_{Z_D} are independent exogenous variables for X , Y , Y_D , Z_C , and Z_D , respectively. The data-generating process in Fig. 1 can be expressed as

$$\begin{aligned} X &= f_X(Z_C, Z_D, \epsilon_X) \\ Y &= f_Y(Z_C, \epsilon_Y) \\ Y_D &= f_{Y_D}(Z_D, \epsilon_{Y_D}) \\ Z_C &= f_{Z_C}(H, \epsilon_{Z_C}) \\ Z_D &= f_{Z_D}(H, D, \epsilon_{Z_D}), \end{aligned} \quad (2)$$

where f_X , f_Y , f_{Y_D} , f_{Z_C} , and f_{Z_D} are the deterministic functions corresponding to X , Y , Y_D , Z_C , and Z_D , respectively. Among them, the function f_X is assumed to be smooth and invertible. It means that

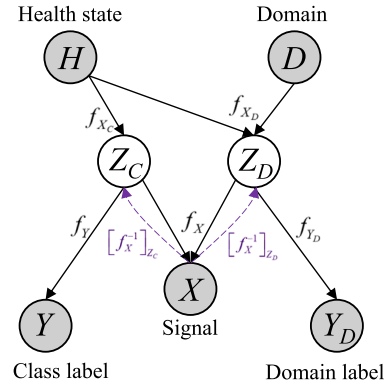


Fig. 1. The structural causal model for fault diagnosis. H is the health state of machine; D is the domain of machine; X is the signal collected from the machine; Y is the class label; Y_D is the domain label; Z_C is the causal factor that is related to machine faults; Z_D is the non-causal factor that is related to domain-specific information. Z_C and Z_D are unobservable variables. f_X , f_Y , f_{Y_D} , f_{Z_C} , and f_{Z_D} are the deterministic functions corresponding to X , Z_C , Z_D , Y , and Y_D . $[f_X^{-1}]_{Z_C}$ and $[f_X^{-1}]_{Z_D}$ are inverse functions of f_X .

$Z_C = [f_X^{-1}]_{Z_C}(X)$ and $Z_D = [f_X^{-1}]_{Z_D}(X)$, where $[f_X^{-1}]_{Z_C}$ and $[f_X^{-1}]_{Z_D}$ are inverse functions of f_X .

From Fig. 1, it can be found that there are two paths connecting the class label Y with the domain D . The first path is $Y \leftarrow Z_C \leftarrow H \rightarrow Z_D \leftarrow D$, which is d-separated because the path is blocked by the node H . However, the other path $Y \leftarrow Z_C \rightarrow X \leftarrow Z_D \leftarrow D$ is d-connected because of the collider (head-to-head) at the node X . It indicates that $Y \not\perp D|X$ (i.e., Y_C is not independent of D given X), which leads to a spurious correlation between the class label Y with the domain D . Therefore, the statistic-based predictor suffers from performance degradation due to domain shift by D . The causality-based DGFD methods can block the spurious correlation by introducing unobservable variables (i.e., Z_C and Z_D), which means that $Y \perp D|Z_C$. In this SCM model, we assume that there are two invariant distributions, i.e., $P(X|Z_C, Z_D)$ and $P(Y|Z_C)$, which are robust to domain shifts. The Z_C and Z_D can be inferred from $P(X|Z_C, Z_D)$ by using a neural network to approximate the $[f_X^{-1}]_{Z_C}$ and $[f_X^{-1}]_{Z_D}$. Once we have Z_C , the causal predictor f_Y , according to that $P(Y|Z_C)$ is invariant cross domains, can be learned in an end-to-end manner.

3. Proposed method

3.1. Problem definition

Let $D_s = \{D_s^i | i = 1, 2, \dots, M\}$ be a training dataset collected from M distinct source domains, where the i th source domain D_s^i contains N_s^i labeled samples, i.e., $D_s^i = \{(\mathbf{x}_j^i, y_j^i) | j = 1, 2, \dots, N_s^i\}$. Each $D_s^i = \{(\mathbf{x}_j^i, y_j^i) | j = 1, 2, \dots, N_s^i\}$ in $D_s = \{(\mathbf{x}_j^i, y_j^i) | j = 1, 2, \dots, N_s^i\}$ is a data-label pair, where $\mathbf{x}_j^i \in \mathcal{X} \subset \mathbb{R}^{N_{input}}$ is the input time-series data, with $\mathcal{X}$ denoting the input space and N_{input} representing the dimension of input space; and $y_j^i \in \mathcal{Y} = \{1, 2, \dots, N_C\}$ is the corresponding category label, with N_C denoting the number of machine health states. Similarly, let $D_t = \{(\mathbf{x}_j^t, y_j^t) | j = 1, 2, \dots, N_t\}$ be test dataset collected from the target domain, where $\mathbf{x}_j^t \in \mathcal{X} \subset \mathbb{R}^{N_{input}}$ and $y_j^t \in \mathcal{Y} = \{1, 2, \dots, N_C\}$. Suppose that each source domain $D_s^i = \{(\mathbf{x}_j^i, y_j^i) | j = 1, 2, \dots, N_s^i\}$ follows the joint distribution P_{XY}^i and the target domain D_t follows P_{XY}^t , then $P_{XY}^1 \neq P_{XY}^2 \neq \dots \neq P_{XY}^M \neq P_{XY}^t$ due to domain shift. The goal of DGFD is to learn a robust mapping $h: \mathcal{X} \rightarrow \mathcal{Y}$ which is only trained on multiple source domains $D_s = \{D_s^i | i = 1, 2, \dots, M\}$ and can generalize well on the unseen target domain D_t .

Let the source domain distribution be p^s and the target domain distribution be p^t . Denote $\mathbb{E}_{p^s}[y|\mathbf{x}] := \int_{\mathcal{Y}} yp^s(y|\mathbf{x})dy$ for any $(\mathbf{x}, y) \in \mathcal{X} \times \mathcal{Y}$ in the source domain, and denote $\mathbb{E}_{p^t}[y|\mathbf{x}] := \int_{\mathcal{Y}} yp^t(y|\mathbf{x})dy$ for

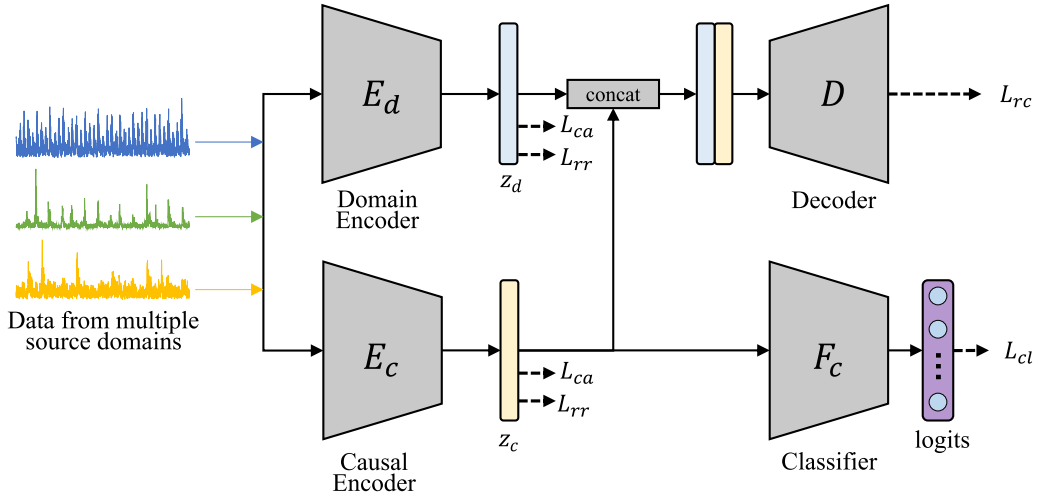


Fig. 2. The diagram of the proposed CDDG.

any $(\mathbf{x}, y) \in \mathcal{X} \times \mathcal{Y}$ in the target domain. Due to the domain-invariance of $P(Y|Z_C)$, we have $\mathbb{E}_{p^s}[y|z_c] = \mathbb{E}_{p^t}[y|z_c] = \mathbb{E}[y|z_c] = \int_{\mathcal{Y}} yp(y|z_c)dy$ and denote $g(Z_C) := \mathbb{E}[Y|Z_C]$. Then the generalization error bound $|\mathbb{E}_{p^s}(y|\mathbf{x}) - \mathbb{E}_{p^t}(y|\mathbf{x})|$ is bounded as follows (Sun et al., 2021):

Theorem 1 (Out of Distribution Generalization Error). Suppose that $p^s(z_c|\mathbf{x})$ and $p^t(z_c|\mathbf{x})$ are probability densities for the source domain and the target domain. $p^s(z_c|\mathbf{x})$ and $p^t(z_c|\mathbf{x})$ are absolutely continuous having support $(-\infty, +\infty)$. For any $(\mathbf{x}, y) \in \mathcal{X} \times \mathcal{Y}$, assume that

- $g(Z_C)$ is a Lipschitz-continuous function;
- $\pi_{\mathbf{x}}(z_c) = \frac{p^t(z_c|\mathbf{x})}{p^s(z_c|\mathbf{x})}$ is differentiable and $\mathbb{E}_{p^s}[\pi_{\mathbf{x}}(Z_C)|g(Z_C) - \mu_1] < \infty$ with $\mu_1 := \mathbb{E}_{p^s}[g(Z_C)|X = \mathbf{x}]$;

Then, we have $|\mathbb{E}_{p^s}(y|\mathbf{x}) - \mathbb{E}_{p^t}(y|\mathbf{x})| \leq \|g'\|_{\infty} \|\pi'_{\mathbf{x}}\|_{\infty} \text{Var}_{p^s}(Z_C|X = \mathbf{x})$, where $\|g'\|_{\infty}$ and $\|\pi'_{\mathbf{x}}\|_{\infty}$ are the Lipschitz constant of g' and π' , respectively. $\text{Var}_{p^s}(Z_C|X = \mathbf{x})$ means the variance of Z_C given $X = \mathbf{x}$ from the source domain.

The theorem describes the error during generalization on the target domain for the strategy that trained on the source domain. Specifically, $\|g'\|_{\infty}$ measures the smoothness of $\mathbb{E}[Y|Z_C]$ with respect to Z_C . $\text{Var}_{p^s}(Z_C|X = \mathbf{x})$ measures the intensity of $\mathbf{x} \rightarrow (z_c, z_d)$. $\|\pi'_{\mathbf{x}}\|_{\infty}$ measures the difference between $p^s(z_c, z_d)$ and $p^t(z_c, z_d)$.

3.2. Overview of CDDG

The diagram of CDDG is shown in Fig. 2. From the figure, it can be seen that CDDG has four parts, including domain encoder E_d , causal encoder E_c , decoder D , and classifier F_c , where E_d , E_c and D are convolutional neural networks (CNNs) and F_c is a multilayer perceptron (MLP) with two fully-connected layers. E_d and E_c share the same structures. In CDDG, the causal factors and non-causal factors are modeled separately. Concretely, the signals $\mathbf{x}$ from multiple source domains are fed into the domain encoder E_d and the causal encoder E_c , then the non-causal factor $\mathbf{z}_d$ and causal factor $\mathbf{z}_c$ are obtained, where $\mathbf{z}_d$ contains the domain-related information and $\mathbf{z}_c$ stores the fault-related information. In order to ensure the causal and non-causal factors are well disentangled, the causal aggregation loss L_{ca} and the redundancy reduction loss L_{rr} are used to regularize $\mathbf{z}_c$ and $\mathbf{z}_d$. Then, $\mathbf{z}_c$ and $\mathbf{z}_d$ are concatenated together and inputted into the decoder D to reconstruct the original signal. The reconstruction loss L_{rc} is applied to constrain the reconstruction process and keep the completeness of the information. Finally, the classifier F_c works as the causal predictor, which takes the causal factor $\mathbf{z}_c$ as input and outputs the logits. The classification loss (i.e., cross-entropy) is used to optimize the F_c .

3.3. Causal aggregation loss

To separate the unobservable causal factor z_c and non-causal factor z_d , the regularization should be imposed on the causal encoder E_c and domain encoder E_d . Intuitively, the causal factors of samples with the same class label should be well aggregated, and the non-causal factors of samples with the same domain label should have a good aggregation. It should be noted that the proposed causal aggregation loss is different from the causal matching loss in Mahajan et al. (2021). First, the method in Mahajan et al. (2021) only tried to decrease the distance between samples with the same label. But in our proposed method, we also consider increasing the distance between samples with different labels. Second, the method in Mahajan et al. (2021) only aggregate causal features. In contrast, our method provides the aggregations of causal and non-causal features simultaneously.

Let n_B be the batch size for each domain, then a batch of samples of M source domains is represented as $\{(\mathbf{x}_j^i, y_j^i) | i = 1, 2, \dots, M, j = 1, 2, \dots, n_B\}$. The data $\{\mathbf{x}_j^i | i = 1, 2, \dots, M, j = 1, 2, \dots, n_B\}$ are passed through the causal encoder E_c and domain encoder E_d , then the causal factor $\{\mathbf{z}_{c_j}^i = E_c(\mathbf{x}_j^i) \in \mathcal{Z}_c \subset \mathbb{R}^{N_d} | i = 1, 2, \dots, M, j = 1, 2, \dots, n_B\}$ and non-causal factor $\{\mathbf{z}_{d_j}^i = E_d(\mathbf{x}_j^i) \in \mathcal{Z}_d \subset \mathbb{R}^{N_d} | i = 1, 2, \dots, M, j = 1, 2, \dots, n_B\}$ are obtained, where N_d is the dimension of the embedding space $\mathcal{Z}_c$ and $\mathcal{Z}_d$. First, the causal aggregation loss of the causal factor $\mathbf{z}_c$ can be calculated by

$$L_{ca}^c = - \frac{\sum_{i_2=1}^M \sum_{j_2=1}^{n_B} \sum_{i_1=1}^M \sum_{j_1=1}^{n_B} 1\{y_{j_1}^{i_1} = y_{j_2}^{i_2}\} \mathbf{z}_{c_{j_1}^{i_1}}^{\top} \mathbf{z}_{c_{j_2}^{i_2}}}{\sum_{i_2=1}^M \sum_{j_2=1}^{n_B} \sum_{i_1=1}^M \sum_{j_1=1}^{n_B} 1\{y_{j_1}^{i_1} = y_{j_2}^{i_2}\}} + \frac{\sum_{i_2=1}^M \sum_{j_2=1}^{n_B} \sum_{i_1=1}^M \sum_{j_1=1}^{n_B} 1\{y_{j_1}^{i_1} \neq y_{j_2}^{i_2}\} \mathbf{z}_{c_{j_1}^{i_1}}^{\top} \mathbf{z}_{c_{j_2}^{i_2}}}{\sum_{i_2=1}^M \sum_{j_2=1}^{n_B} \sum_{i_1=1}^M \sum_{j_1=1}^{n_B} 1\{y_{j_1}^{i_1} \neq y_{j_2}^{i_2}\}}. \quad (3)$$

Similarly, the causal aggregation loss of the non-causal factor $\mathbf{z}_d$ can be expressed as

$$L_{ca}^d = - \frac{\sum_{j_2=1}^{n_B} \sum_{j_1=1}^{n_B} \sum_{i=1}^M \mathbf{z}_{d_{j_1}^i}^{\top} \mathbf{z}_{d_{j_2}^i}}{M n_B^2} + \frac{\sum_{i_2=1}^M \sum_{j_2=1}^{n_B} \sum_{i_1=1}^M \sum_{j_1=1}^{n_B} 1\{i_1 \neq i_2\} \mathbf{z}_{d_{j_1}^{i_1}}^{\top} \mathbf{z}_{d_{j_2}^{i_2}}}{(M-1)M n_B^2}. \quad (4)$$

Thus, the causal aggregation loss is computed by

$$L_{ca} = L_{ca}^c + L_{ca}^d. \quad (5)$$

3.4. Reconstruction loss

Although the casual aggregation loss can help disentangle causal and non-causal factors, trivial solutions (e.g., $\mathbf{z}_c = \mathbf{0}$ or $\mathbf{z}_d = \mathbf{0}$) may be attained, and certain useful information may lose as a result. To avoid trivial solutions and to reduce information loss, reconstruction loss L_{rc} is introduced for enhancing feature learning. The causal factor $\{\mathbf{z}_{c_j}^i = E_c(\mathbf{x}_j^i) \in \mathcal{Z}_c \subset \mathbb{R}^{N_d} | i = 1, 2, \dots, M, j = 1, 2, \dots, n_B\}$ and the non-causal factor $\{\mathbf{z}_{d_j}^i = E_d(\mathbf{x}_j^i) \in \mathcal{Z}_d \subset \mathbb{R}^{N_d} | i = 1, 2, \dots, M, j = 1, 2, \dots, n_B\}$ outputted by the causal encoder E_c and the domain encoder E_d are concatenated. They are then decoded by the decoder D to reconstruct the raw input. Let $\{\hat{\mathbf{x}}_j^i = D([\mathbf{z}_{c_j}^i, \mathbf{z}_{d_j}^i]) | i = 1, 2, \dots, M, j = 1, 2, \dots, n_B\}$ denote the reconstructed input signals by D . The reconstruction loss can be written as

$$L_{rc} = \frac{1}{M n_B N_{input}} \sum_{j=1}^M \sum_{i=1}^{n_B} \|\mathbf{x}_j^i - \hat{\mathbf{x}}_j^i\|_2^2. \quad (6)$$

3.5. Redundancy reduction loss

The causal and non-causal factors are regularized by L_{ca} and L_{rc} , but there are still two limitations. The first one is the possible correlation between the feature dimensions of factor vectors. According to Zhang, Cui, et al. (2021), the correlation between features may have a negative impact on model prediction. The second one is the entanglement between causal and non-causal factors, which would affect the domain-invariance of causal factors.

Inspired by the redundancy-reduction principle in Barlow Twins (Zbontar et al., 2021), the redundancy reduction loss is designed to regularize causal and non-causal factors. Let $\mathbf{Z}_c \in \mathbb{R}^{M n_B \times N_d}$ denote the causal feature matrix, where each row represents the causal factor $\mathbf{z}_{c_j}^i$. Similarly, the non-causal feature matrix can be represented by $\mathbf{Z}_d \in \mathbb{R}^{M n_B \times N_d}$, and each row corresponds to the non-causal factor $\mathbf{z}_{d_j}^i$. Then, the redundancy reduction loss can be expressed as

$$L_{rr} = \frac{\|(\mathbf{I} - \mathbf{E}) \odot (\mathbf{Z}_c^T \mathbf{Z}_c)\|_F^2}{N_d(N_d - 1)} + \frac{\|(\mathbf{I} - \mathbf{E}) \odot (\mathbf{Z}_d^T \mathbf{Z}_d)\|_F^2}{N_d(N_d - 1)} + \frac{\|\mathbf{Z}_c^T \mathbf{Z}_d\|_F^2}{N_d^2}, \quad (7)$$

where $\mathbf{I} \in \mathbb{R}^{N_d \times N_d}$ is a square matrix with the element of 1; $\mathbf{E} \in \mathbb{R}^{N_d \times N_d}$ is the identity matrix; $\odot$ means Hadamard product; $\|\cdot\|_F$ means the Frobenius norm.

3.6. Classification loss

The causal factor $\mathbf{z}_c$ contains sufficient fault-related information for fault classification. In order to learn the causal predictor (i.e., classifier F_c), the cross-entropy is used as the classification loss. After fed in the causal factor $\mathbf{z}_{c_j}^i$, the logit $\hat{\mathbf{y}}_j^i = F(\mathbf{z}_{c_j}^i) \in \mathbb{R}^{N_c}$ outputted by F_c is obtained with the SoftMax activation. Suppose that $\mathbf{y}_j^i$ is the one-hot code of the label y_j^i , then the classification loss is

$$L_{cl} = - \sum_{j=1}^M \sum_{i=1}^{n_B} \mathbf{y}_j^i{}^T \log \hat{\mathbf{y}}_j^i. \quad (8)$$

3.7. Overall framework

Combining the causal aggregation loss, reconstruction loss, redundancy reduction loss and classification loss, the overall optimization objective of CDDG is

$$L = L_{cl} + \alpha L_{ca} + \beta L_{rc} + \gamma L_{rr}, \quad (9)$$

where α , β , and γ are the hyperparameters that balance the regularization items.

Let θ_{E_c} , θ_{E_d} , θ_D , and θ_{F_c} denote the parameters of E_c , E_d , D , and F_c , respectively. The parameter updating of each part of CDDG at t th iteration, that is,

$$\begin{aligned} \theta_{E_c}^{(t+1)} &\leftarrow \theta_{E_c}^{(t)} - \eta^{(t)} \left(\frac{\partial L_{cl}}{\partial \theta_{E_c}} + \alpha \frac{\partial L_{ca}}{\partial \theta_{E_c}} + \beta \frac{\partial L_{rc}}{\partial \theta_{E_c}} + \gamma \frac{\partial L_{rr}}{\partial \theta_{E_c}} \right) \\ \theta_{E_d}^{(t+1)} &\leftarrow \theta_{E_d}^{(t)} - \eta^{(t)} \left(\alpha \frac{\partial L_{ca}}{\partial \theta_{E_d}} + \beta \frac{\partial L_{rc}}{\partial \theta_{E_d}} + \gamma \frac{\partial L_{rr}}{\partial \theta_{E_d}} \right) \\ \theta_D^{(t+1)} &\leftarrow \theta_D^{(t)} - \eta^{(t)} \gamma \frac{\partial L_{rr}}{\partial \theta_D} \\ \theta_{F_c}^{(t+1)} &\leftarrow \theta_{F_c}^{(t)} - \eta^{(t)} \gamma \frac{\partial L_{cl}}{\partial \theta_{F_c}}, \end{aligned} \quad (10)$$

where $\eta^{(t)}$ is the learning rate at the t th iteration. The algorithm of CDDG framework is given in Algorithm 1.

Algorithm 1 Algorithm for CDDG.

Training Stage:

Require: Training data D_s from M domains, the batch size n_B , the learning rate δ , the number of fault types N_c , the number of training steps n_{step}

Module: The untrained network E_c , E_d , D , and F_c

Return: The trained networks E_c , E_d , D , and F_c

- 1: **Initialization:** θ_{E_c} , θ_{E_d} , θ_D , and θ_{F_c} ▷ kaiming-normal
 - 2: **for** $t = 1, 2, \dots, n_{step}$ **do**
 - 3: 1) Randomly sample a batch of data $\{(\mathbf{x}_j^i, y_j^i) | i = 1, 2, \dots, M, j = 1, 2, \dots, n_B\}$ from D_s
 - 4: 2) Forward propagation of the network
 - 5: Calculate the causal aggregation loss L_{ca} by Eq. (3)–Eq. (5)
 - 6: Calculate the reconstruction loss L_{rc} by Eq. (6)
 - 7: Calculate the redundancy reduction loss by Eq. (7)
 - 8: Calculate the classification loss L_{cl} by Eq. (8)
 - 9: 3) Backpropagation
 - 10: Update the parameters of E_c , E_d , D , and F_c by Eq. (10)
 - 11: **end for**
-

Test Stage:

Require: Test data D_t from the target domain

Module: The trained networks E_c and F_c in the training stage

Return: The predicted labels y^t

- 1: **for** each $\mathbf{x}_j^t$ in $D_t = \{\mathbf{x}_j^t | j = 1, 2, \dots, N_t\}$ **do**
 - 2: $y_j^t = \arg \max(F_c(E_c(\mathbf{x}_j^t)))$
 - 3: **end for**
-

4. Experiments

In this section, the CDDG will be verified on cross-machine vibrational bearing fault diagnosis tasks and cross-environment acoustical fan anomaly detection tasks. Both vibrational signals and acoustical signals are representative time-series data commonly used in fault diagnosis field. Extensive experiments including comparative analysis, ablation study, parameter sensitivity analysis, and visualization, are conducted to demonstrate the superiority of the proposed method.

4.1. Case study 1: Cross-machine bearing fault diagnosis

4.1.1. Dataset description

In this case study, experimental validation of cross-machine domain generalization utilizing five distinct bearing fault time-series datasets is presented. As reported in the literature (Jia, Li, et al., 2023), a majority of bearing failures are attributed to faults in either the inner races (IR) or outer races (OR). Accordingly, only these two types of faults

Table 1

The detailed information of the five datasets.

Code	Dataset	Bearing types	Speeds (Hz)	Loads	Sampling rate (Hz)	Number of training samples	Number of test samples
A	CWRU	6205-RS JEM SKF	[29.95,29.53,29.17,28.83]	[0.1,2,3] hp	12 000	840	840
B	MFS	ER16K	[3:0.5:9] + [10:1:15]	[49] N	10 240	6270	2280
C	UOTTAWA	ER16K	time-varying	–	200 000	360	360
D	MFPT	–	[25]	[25]+[50:50:300] lbs.	48 828	150	150
E	DZLRB	FAGQJ212TVP	[8.33,1]	[5000,8800] N	25 600	840	840

are considered and included in the fault classification scheme, which comprises three categories, namely normal (NM), IR fault, and OR fault.

(1) CWRU Dataset (Smith & Randall, 2015): The CWRU dataset is provided by Case Western Reserve University and is widely recognized as a prominent benchmark for bearings fault diagnosis. The dataset incorporates vibrational signals that were acquired from both the drive-end and the fan-end of a motor, subjected to four different loads (0hp, 1hp, 2hp, 3hp), with a sampling frequency of 12000 Hz. In this paper, only the drive-end signals have been utilized.

(2) MFS Dataset (Li, Wang, Zi, & Zhang, 2021): The MFS dataset was provided by Xi'an Jiaotong University. It contains vibrational signals acquired from a mechanical fault simulation test rig propelled by a single-phase asynchronous motor operating under a constant load of 49N. The experimental procedure involved running the machine for 48 s, while varying the rotational speeds within a range of 3 Hz to 9 Hz, with a step size of 0.5 Hz (i.e., 3, 3.5, ..., 9.0 Hz) and within a range of 10 Hz to 15 Hz with a step size of 1 Hz (i.e., 10, 11, ..., 15 Hz). The dataset's sampling frequency is 10240 Hz.

(3) UOTTAWA Dataset (Huang & Baddour, 2018): The UOTTAWA dataset, published by the University of Ottawa, consists of vibrational signals obtained from a Spectra-Quest machinery fault simulator. The dataset's operational conditions entail varying rotational speeds, including increasing speed, decreasing speed, increasing followed by decreasing speed, and decreasing followed by increasing speed. The sampling frequency for the dataset is 200000 Hz.

(4) MFPT Dataset (Bechhoefer, 2016): The MFPT dataset, which was offered by the Society for Machinery Failure Prevention Technology, comprises vibrational signals obtained from a test rig operating under various loads of 25, 50, 100, 150, 200, 250, and 300 lbs. The dataset's sampling rate is 48828 Hz, while the machine has a constant rotational rate of 25 Hz.

(5) DZLRB Dataset (Ismail, Bierig, & Sawalhi, 2018): The DZLRB dataset is from Deutsches Zentrum für Luft und Raumfahrt Standort Braunschweig. The experimental setup involved a test rig operating under an axial load of 5kN and 8.8kN, with two rotational speeds of 8.33 Hz and 1 Hz. The dataset encompasses fault sizes of 1.0 mm, 2.1 mm, and 3.8 mm for the inner race fault type, and fault sizes of 1.4 mm, 2.4 mm, and 4.0 mm for the outer race fault type. The sampling frequency utilized for data acquisition is 25600 Hz.

The details of the five bearing datasets are illustrated in Table 1. Given the differences in sampling frequencies and rotating speeds among the five bearing fault datasets, appropriate signal preprocessing steps are deemed necessary. In line with prior research (Zheng et al., 2020), the current study employs angular resampling and Hilbert transform for this purpose. The resulting signal envelopes, obtained following the Hilbert transform, are subsequently utilized as inputs to the neural network model for fault classification.

4.1.2. Implementation details

The structural parameters of causal/domain encoder, decoder, and classifier are listed in Table 2. The data length is 2560. The batch size n_B is 64, and the model is trained 200 iterations. The initial learning rate is set to 0.0001. In order to stabilize the training process, the learning rate decay strategy is adopted. The learning rate is reduced by a

Table 2

The structural parameter of the network of CDDG for cross-machine bearing fault diagnosis.

	Layer	Parameter	Activation	Output shape
Causal Domain Encoder	Conv1D+BN	(1, 32, 128)	LReLU	(B, 32, 2433)
	MaxPool1d	2	–	(B, 32, 1216)
	Conv1D+BN	(32, 32, 64)	LReLU	(B, 32, 1153)
	MaxPool1d	2	–	(B, 32, 576)
	Conv1D+BN	(32, 32, 32)	LReLU	(B, 32, 545)
	MaxPool1d	2	–	(B, 32, 272)
	Conv1D+BN	(32, 32, 16)	LReLU	(B, 32, 257)
	MaxPool1d	2	–	(B, 32, 128)
	Conv1D+BN	(32, 32, 5)	LReLU	(B, 32, 124)
	MaxPool1d	2	–	(B, 32, 62)
	Flatten	–	–	(B, 1984)
	Upsample	2	–	(B, 64, 124)
Decoder	Conv1D+BN	(64, 32, 5)	LeakyReLU	(B, 32, 128)
	Upsample	2	–	(B, 32, 256)
	Conv1D+BN	(32,32,15)	LeakyReLU	(B, 32, 272)
	Upsample	2	–	(B, 32, 544)
	Conv1D+BN	(32,32,31)	LeakyReLU	(B, 32, 576)
	Upsample	2	–	(B, 32, 1152)
	Conv1D+BN	(32,32,63)	LeakyReLU	(B, 32, 1216)
	Upsample	2	–	(B, 32, 2432)
	Conv1D+BN	(32,32,127)	LeakyReLU	(B, 32, 2560)
	Conv1D+BN	(32,1, 127)	–	(B, 1, 2560)
Classifier	Linear	(1984, 300)	–	(B, 300)
	Linear	(300, 3)	SoftMax	(B, 3)

factor of 0.1 after every 50 iterations. Adam is selected as the optimizer. The hyperparameter α , β , and γ are set to 1, 1, 0.1, respectively. In this experimental setting, we follow the leave-one-domain-out rule, it means that one domain is considered as the target domain and the other four domains are regarded as the source domain, i.e., $M = 4$. The CDDG model is implemented with PyTorch 1.9.0 and runs on the computer equipped with Nvidia RTX 2080Ti GPU.

4.1.3. Comparative analysis

To validate the effectiveness of the proposed CDDG, eight popular SOTA DGFD methods, including CNN-C (Yang et al., 2020), DG-NIS (Zhao & Shen, 2022), CCDG (Ragab et al., 2022), DDGFD (Zheng et al., 2020), IEDGNet (Han et al., 2021), ADN (Li et al., 2020a), Whitening-Net (Li, Wang, Zi, & Zhang, 2021), and CCN (Li et al., 2022), are implemented for comparison. Among the above SOTA methods, the Whitening-Net (Li, Wang, Zi, & Zhang, 2021) and CCN (Li et al., 2022) are causality-based DGFD methods. For a fair comparison, all the methods adopt the same network structures as CCDG. Each case is repeated five times to reduce randomness. The average value and standard deviation of accuracies are recorded, and the results are provided in Table 3. It can be seen that the proposed CDDG achieved an overall accuracy of 81.23% showing the best performance among all comparative methods. CCN outperformed other SOTA methods with an average accuracy of 77.50% because it can mine the invariant causal information by emphasizing the consistency of latent variables and predictions. By comparison, the other causality-based method, Whitening-Net, only reached 67.31%, which is 10.19% lower than CCN because Whitening-Net only considered the causal variable. Different from CNN and Whitening-Net, our CDDG regularizes the causal

Table 3

The comparative results of the eight SOTA methods, the proposed CDDG and its ablated models on cross-machine bearing fault diagnosis cases. (Note that “BCDE→A” in the header means the dataset B, C, D, and E are source domains and the dataset A is the target domain. It means similar to other cases.)

	BCDE→A	ACDE→B	ABDE→C	ABCE→D	ABCD→E	AVER.
CNN-C	70.97 ± 1.63	82.91 ± 2.37	63.45 ± 8.80	65.33 ± 2.26	46.88 ± 7.31	65.91
DGNIS	80.14 ± 2.76	75.55 ± 3.64	62.72 ± 3.54	74.53 ± 6.38	53.93±11.05	69.38
CCDG	78.71 ± 3.19	84.84±1.78	66.81 ± 1.71	79.47 ± 7.75	52.88 ± 7.78	72.54
DDGFD	80.19 ± 3.21	84.10 ± 1.44	63.72 ± 1.07	72.80 ± 4.48	51.98 ± 5.43	70.56
IEDGNet	80.86 ± 1.93	78.73 ± 5.59	56.67 ± 5.08	82.13 ± 6.62	50.40 ± 2.93	69.76
ADN	76.40 ± 4.83	81.75 ± 3.22	72.74 ± 7.08	86.00 ± 9.60	49.10 ± 5.97	73.20
Whitening-Net	74.17 ± 3.23	83.74 ± 0.97	58.78 ± 2.39	74.40 ± 7.65	45.48 ± 6.12	67.31
CCN	83.52 ± 2.26	78.74 ± 2.98	84.89 ± 5.44	88.13 ± 2.18	52.23 ± 8.01	77.50
CDDG _{no_ca}	84.74 ± 1.24	80.39 ± 3.83	89.56 ± 3.39	86.53 ± 6.55	43.31 ± 15.36	76.90
CDDG _{no_rc}	85.14 ± 0.83	80.66 ± 3.05	87.78 ± 2.87	90.67 ± 9.92	41.88 ± 4.84	77.22
CDDG _{no_rr}	85.36 ± 0.50	81.98 ± 3.39	91.94 ± 1.87	89.60 ± 7.16	41.97 ± 3.95	78.17
CDDG	85.50±0.62	82.34 ± 2.44	92.05±3.18	94.80±3.93	53.00 ± 2.54	81.54

and non-causal factors simultaneously, and the spurious correlations in causal factors are reduced significantly. CNN-C only obtained an average accuracy of 65.91%, which is superior to other SOTA methods. Especially, DGNIS and CCDG achieved the best result in the case ABCD→E (i.e., the dataset A, B, C, and D are source domains, and the dataset E is the target domain) and the case ACDE→B, respectively. However, there are large gaps between them and CCDG in the other three individual cases. For example, CCDG is 25.24% lower than the proposed CDDG in the case ABDE→C. The average accuracy of 70.56% for DDGFD indicated that only imposing a penalty on samples near the decision boundary may mislead the model. IEDGNet and ADN are both adversarial-based DGFD methods, the difference between them is that ADN enhances consistency via deep metric learning, which makes ADN the second-highest overall accuracy in the SOTA methods.

In order to verify the effectiveness of each regularization loss, ablation experiments are carried out. We remove L_{ca} , L_{rc} , and L_{rr} from CDDG, then three ablated models including CDDG_{no_ca}, CDDG_{no_rc}, and CDDG_{no_rr} are obtained. The accuracy results of the ablated models are all listed in Table 3. From the table, it is evident that the removal of any of the three losses would lead to performance degradation. The CDDG has the highest average accuracy, and it surpasses the CDDG_{no_ca}, CDDG_{no_rc}, and CDDG_{no_rr} by 4.64%, 4.32%, and 3.37%, respectively. The three regularization losses have a positive effect on the all of the five cases. Thus, the results confirm that the proposed causal aggregation loss, reconstruction loss, and redundancy reduction loss are useful to the model to learn stable causal features and generalized diagnostic knowledge.

4.1.4. Hyperparameter sensitivity analysis

The performance of CDDG is heavily related to the selection of hyperparameters α , β , and γ , which are corresponding to the weights of causal aggregation loss, reconstruction loss, and the redundancy reduction loss, respectively. To investigate the sensitivity of the hyperparameters to the model performance, extensive experiments are conducted. We test the three hyperparameters of CDDG on five cases and run each case by performing grid search in the set [0.01, 0.1, 1, 5, 10]. The results of sensitivity analysis are illustrated in Fig. 3. First, it can be seen that the CDDG model is more sensitive to γ than α and β . The average accuracy decreases when γ is larger than 0.1. It means that the redundancy reduction regularization should not be very strong. The possible reason is that some redundancy information is highly related to model generalization. Second, the choice of α does not have a significant effect on most cases except the case ABCE→D. CDDG obtains the highest accuracy when $\alpha = 1$. Finally, different cases show various pattern with the change of β . Among the five cases, the model will degrade the case BCDE→A and ABDE→C if β is too large. In contrast, the increase of β brings a positive effect on accuracy for case ABCD→E. CDDG performs best when β is set to 1. Therefore, the trade-off hyperparameters α , β , and γ are 1, 1, and 0.1, respectively.

4.1.5. Visualization

In order to evaluate the classification performance of the CDDG on each class, confusion matrices for the five DGFD cases are presented in Fig. 4. From the figure, it can be seen that the CDDG can accurately differentiate the IR fault of the CWRU dataset on the case BCDE→A, but some OR samples are misclassified as NM. In the case of case ACDE→B, almost all of the faulty samples in the MFS dataset are correctly identified as abnormal, although about 384 IR samples are mislabeled as OR. When the UOTTAWA dataset is used as the target domain, CCDG can discriminate between NM and IR samples, but 19 OR samples are erroneously classified as NM. MFPT is the most easily classified, with all classes being well classified. Specially, in the case ABCD→E, CDDG exhibits the poorest performance, and all OR samples are misclassified as IR. As a whole, the normal state can be recognized accurately and robustly; thus the false alarming rate is very low that is important in practical engineering.

To further verify the causal feature learning ability of CDDG, the t-SNE is used for visualization, as shown in Fig. 5. From the five t-SNE subfigures, we can see that the NM class always clusters well, and the OR fault demonstrates a relatively poor clustering performance. Meanwhile, the clustering of the OR class in Fig. 5(d) is better than the other four cases, and the samples of target domain are distant from class boundaries. It is consistent with the observation in the confusion matrix result in Fig. 4(d). As for the case ABCD→E, it is noticed that the OR samples of the DZLRSB dataset are mapped near the IR samples of MFPT, this may explain why all OR samples of DZLRSB are misclassified.

The domain shifts across different domains are associated with variations in the non-causal factor, which contains domain-specific features. Therefore, it is necessary to visualize the non-causal factor to show differences between different domains. Here we use principal component analysis (PCA) to reduce the dimensions of feature vectors of each domain, and the first two principal components of the five cases are plotted, as shown in Fig. 6 (domain label) and Fig. 7 (health condition label). In Fig. 6, each color of a sample point represents one specific domain. First, it can be seen that the position of each domain is relatively fixed whether the domain works as the source domain or the target domain for all five cases. It means that the proposed CDDG has a stable performance. Second, there is almost no overlap between domain A (red), domain B (green), and domain C (blue), and it indicates that the non-causal factor contains huge domain-specific information. Meanwhile, there is no significant clustering in the visualization. The reason may be that the non-causal factor contains not only domain-specific information, but also some common features. The common features are unrelated to the domain label or the health condition label but are necessary for signal reconstruction. In Fig. 7, each color corresponds to a specific machine health condition. It can be observed that the IR fault and the OR fault are indistinguishable for all five cases. Therefore, there is little fault-related information in z_d . However, the samples of the NM condition are clustered well and

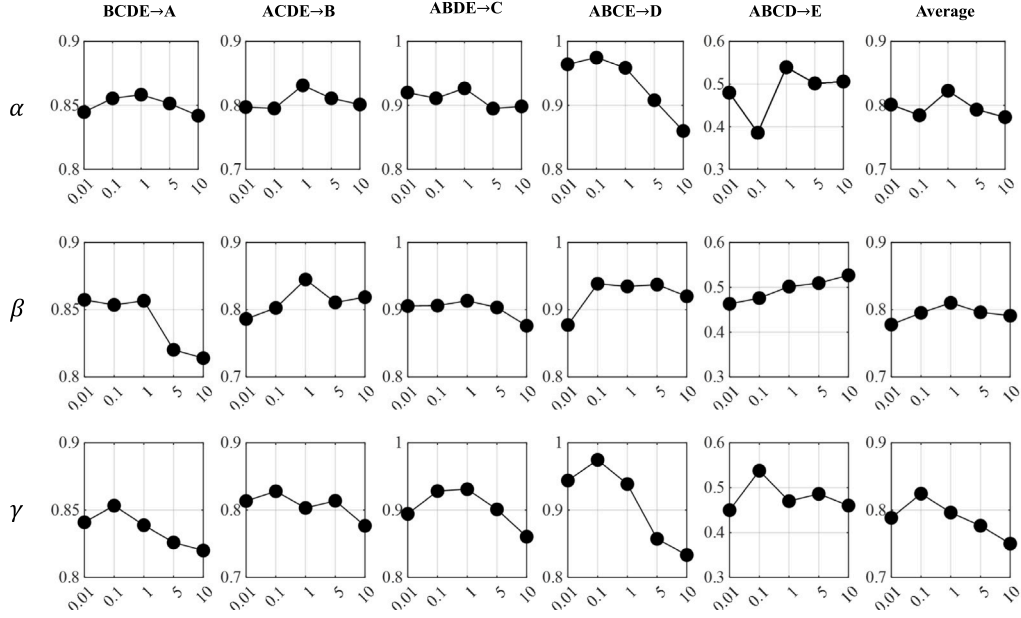


Fig. 3. Hyperparameter sensitivity analysis on five cases. The first row represents the sensitivity of α ; The second row represents the sensitivity of β ; The third row represents the sensitivity of γ .

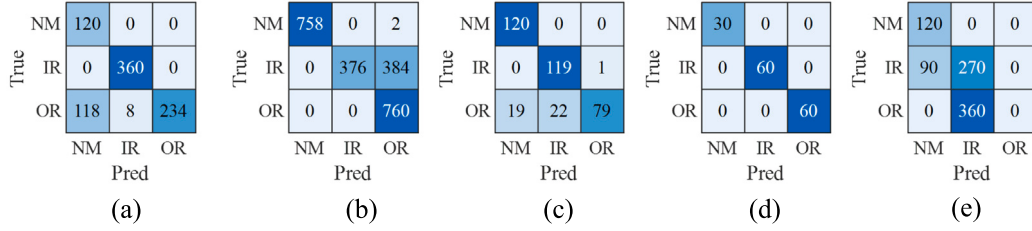


Fig. 4. The confusion matrices of CDDG. (a) case BCDE→A; (b) case ACDE→B; (c) case ABDE→C; (d) case ABCE→D; (e) case ABCD→E.

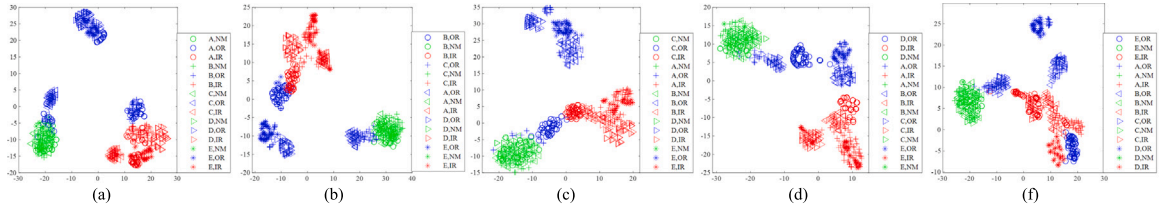


Fig. 5. The t-SNE visualizations of the penultimate layer of CDDG classifier. (a) case BCDE→A; (b) case ACDE→B; (c) case ABDE→C; (d) case ABCE→D; (e) case ABCD→E.

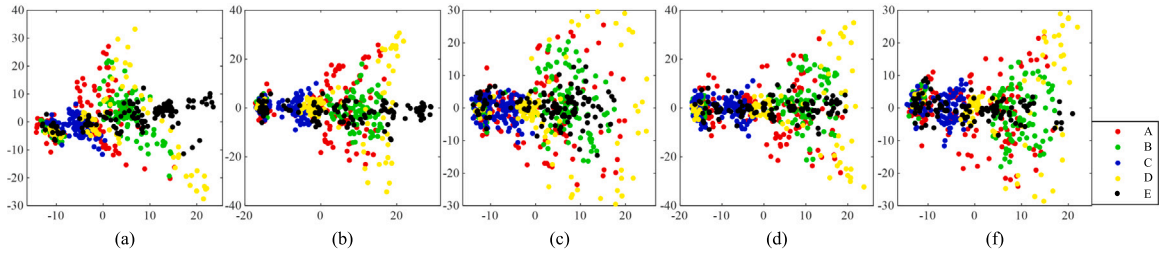


Fig. 6. The PCA visualizations of the non-causal factor z_d with the domain label. (a) case BCDE→A; (b) case ACDE→B; (c) case ABDE→C; (d) case ABCE→D; (e) case ABCD→E.

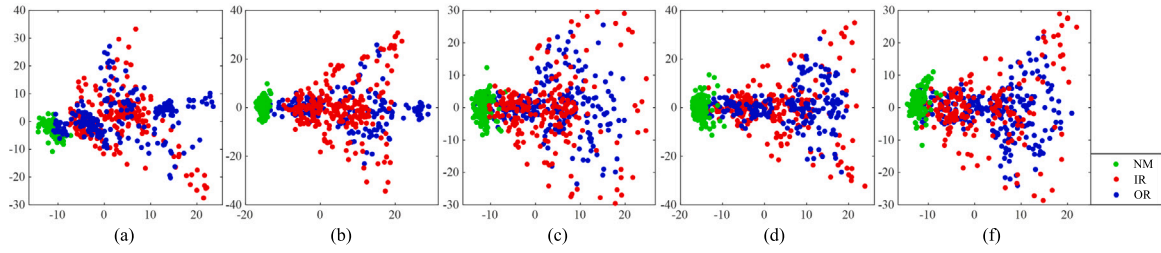


Fig. 7. The PCA visualizations of the non-causal factor z_d with health condition label. (a) case BCDE→A; (b) case ACDE→B; (c) case ABDE→C; (d) case ABCE→D; (e) case ABCD→E.

Table 4

The detailed information of the fan part of the MIMII DG dataset.

	Source domains	Target domain	Description
Section 00	X, Y, Z	W	Mixing of different machine sound
Section 01	B, C	A	Mixing of different factory noise
Section 02	L2, L3, L4	L1	Different levelsof noise

almost have no overlap with the other two health conditions (i.e., IR and OR). This is because the normal condition samples from different domains have similar distributions and show a very different pattern from the samples of fault conditions. The causal factor z_c captures some fault-related information that is enough for fault recognition, but the non-causal factor z_d keeps the rest of the fault-related information for reconstruction.

4.2. Case study 2: Cross-environment fan anomaly detection

4.2.1. Dataset description

To further verify the application potential of the proposed CDDG, the cross-environment acoustical fan anomaly detection is explored in this section. We use the MIMII DG sound dataset (Dohi et al., 2022) collected by Hitachi Ltd. This dataset contains sound signals acquired from five machine types including fan, gearbox, bearing, sliding rail, and valve. In this study, only the fan dataset is used. The fan dataset has three sections (i.e., section 00, section 01, and section 02), and each section consists of three or four domains, as listed in Table 4. Each sample in MIMII DG is a 10-second sound signal with a sampling frequency of 16000 Hz. For convenience, three DGFD cases, including XYZ→W, BC→A, and L2L3L4→L1, are designed for experiments.

4.2.2. Implementation details

The length of each sound signal in the dataset is 160000, rendering it unsuitable for direct processing by CNN. Therefore, preprocessing of the data is required. In accordance with Dohi et al. (2022), we initially transform the 1D sound signal into a 2D Mel-Spectrum representation, utilizing a frame size of 1024, a hop size of 210, and a Mel bin number of 128. Subsequently, we flatten the 2D Mel-Spectrum to generate a 1D signal with a data length of 20032. In order to fit the new data length, a new network structure is designed, as provided in Table 5. The batch size is set to 32. The initial learning rate is 0.0001, and the learning rate halves at the 25-th iteration. Other settings are the same as the former case study.

4.2.3. Comparative analysis

Eight SOTA DGFD methods, including CNN-C (Yang et al., 2020), DGNIS (Zhao & Shen, 2022), CCDG (Ragab et al., 2022), DDGFD (Zheng et al., 2020), IEDGNet (Han et al., 2021), ADN (Li et al., 2020a), Whitening-Net (Li, Wang, Zi, & Zhang, 2021), and CCN (Li et al., 2022), are used for comparison. The comparison results are reported in Table 6. From the table, it is clear that the proposed CDDG outperforms the eight methods in all individual cases and achieves the best overall average accuracy of 78.68%, which surpasses the second-best method (i.e., DDGFD) by 4.12%. Similar to the first case study, the ADN and

CCN show a relatively high ranking, while the CNN-C and Whitening-Net demonstrate relative underperformances in the comparative methods. Across the three DGFD cases, CDDG exhibits the largest gap in performance in the case XYZ→W compared to the best SOTA method. In summary, the proposed CDDG is suitable for cross-environment acoustical fan anomaly detection tasks and it exhibits superiority in causal invariant feature disentanglement. The results suggest that CDDG can acquire generalized knowledge from multiple source domains. Thus, it can be seen that the proposed CDDG can contribute significant advantages in real-world fault diagnosis scenarios.

5. Limitations and future works

Although the proposed method has achieved a good performance, there are still some limitations.

(1) Quantity of dataset: The model training process needs a large quantity of training samples. We can analyze dataset quantity from two aspects: size and diversity. First, some datasets have very limited sizes. A solution for this issue is to use data augmentation method (e.g., mix-up (Zhang, Cisse, Dauphin, & Lopez-Paz, 2017), GAN (Goodfellow et al., 2020)) to generate more samples. Second, to improve the dataset diversity, we may use datasets from many sources, including open-source datasets, numerical simulation datasets, and private datasets. In addition, many data owners do not share their datasets because of commercial interests. Thus, developing federated learning DG models (Zhao & Shen, 2023a) is also a promising solution for enhancing dataset quantity.

(2) Quality of dataset: The quality of datasets can determine the model performance to a great extent. Some datasets are corrupted by noises, and the fault characteristics are not discriminative. In order to solving this problem, a possible solution is to use some backbones (e.g., Li, Li, Sun, Chen, & Yan, 2023) that is robust to noises or some network modules for signal preprocessing (e.g., Wang, Liu, Peng, & Cheng, 2022).

(3) Identifiability on target domain: Whether the causal factor Z_C can be recovered from the non-causal factor Z_D is an identifiability problem. If the causal factor can be identified, it means that properties of the underlying data generating process can be uncovered (Von Kügelgen, Sharma, Gesele, Brendel, Schölkopf, Besserve, & Locatello, 2021). Liu et al. (2021) and Sun et al. (2021) have theoretically proven that the identifiability of the causal factor is possible in single-source and multi-source domains. However, they only showed that causal factor identification is feasible in source domains. Whether the causal factor can be identified in the unseen target domain is still an open problem.

Considering the limitations, in the future work, we will first explore causal-based data augmentation methods for federated learning domain

Table 5

The structural parameter of the network of CDDG for cross-environment fan anomaly detection.

	Layer	Parameter	Activation	Output shape
Feature extractor	Conv1D+BN	(1, 32, 128)	LReLU	(B, 32, 19905)
	MaxPool1d	4	–	(B, 32, 4976)
	Conv1D+BN	(32, 32, 128)	LReLU	(B, 32, 4849)
	MaxPool1d	4	–	(B, 32, 1212)
	Conv1D+BN	(32, 32, 128)	LReLU	(B, 32, 1085)
	MaxPool1d	4	–	(B, 32, 271)
	Conv1D+BN	(32, 32, 64)	LReLU	(B, 32, 208)
	MaxPool1d	2	–	(B, 32, 104)
	Conv1D+BN	(32, 32, 64)	LReLU	(B, 32, 41)
	MaxPool1d	2	–	(B, 32, 20)
Decoder	Flatten	–	–	(B, 640)
	Upsample	2	–	(B, 64, 40)
	Conv1D+BN	(64, 32, 64)	LeakyReLU	(B, 32, 103)
	Upsample	2	–	(B, 32, 206)
	Conv1D+BN	(32, 32, 64)	LeakyReLU	(B, 32, 271)
	Upsample	4	–	(B, 32, 1084)
	Conv1D+BN	(32, 32, 128)	LeakyReLU	(B, 32, 1211)
	Upsample	4	–	(B, 32, 4844)
	Conv1D+BN	(32, 32, 128)	LeakyReLU	(B, 32, 4975)
	Upsample	4	–	(B, 32, 19900)
Classifier	Conv1D+BN	(32, 32, 128)	LeakyReLU	(B, 32, 20031)
	Conv1D+BN	(32, 1, 128)	–	(B, 32, 20032)
Classifier	Linear	(640, 300)	–	(B, 300)
	Linear	(300, 2)	SoftMax	(B, 2)

Table 6

The comparative results of the proposed CDDG and the eight SOTA DGFD methods on cross-environmental fan anomaly detection cases. (Note that “XYZ→W” in the header means the dataset X, Y, and Z are source domains and the dataset W is the target domain. It means similar to other cases.)

	XYZ→W	BC→A	L2L3L4→L1	AVER.
CNN-C	84.44 ± 3.51	54.69 ± 0.91	68.57 ± 6.39	69.23
DGNIS	82.67 ± 3.98	53.47 ± 7.96	78.57 ± 4.45	71.57
CCDG	84.00 ± 8.37	51.73 ± 6.36	81.43 ± 3.48	72.39
DDGFD	85.78 ± 5.58	57.55 ± 7.96	80.36 ± 2.18	74.56
IEDGNet	84.00 ± 13.19	50.61 ± 2.66	72.50 ± 7.21	69.04
ADN	84.49 ± 7.76	56.33 ± 7.85	80.71 ± 4.07	73.84
Whitening-Net	79.11 ± 3.37	55.51 ± 0.91	65.72 ± 10.89	66.78
CCN	77.78 ± 6.66	58.78 ± 4.19	83.93 ± 5.21	73.50
CDDG	90.22±2.53	60.81±2.24	85.00±2.04	78.68

generalization. Second, we will investigate how to use normal samples from the target domain to further improve the model performance.

6. Conclusion

This article proposed a novel domain generalization-based time-series signal fault diagnosis method called CDDG. In contrast to existing statistical models, CDDG offers a unique perspective by approaching the DGFD problem from the viewpoint of causal learning through the reestablishment of the data-generating process in SCM form. The design of the causal encoder and domain encoder within CDDG ensures the disentanglement of fault-related causal factors and domain-related non-causal factors, respectively. The causal aggregation loss is utilized to encourage the grouping together of samples with the same health state label and the same domain label. Additionally, the reconstruction loss prevents trivial solutions while ensuring information completeness. The redundancy reduction loss helps suppress redundancy in learned feature representations and mitigate spurious correlations between causal and non-causal factors. Extensive experiments were conducted on five cross-machine vibrational bearing fault diagnosis cases and three cross-environmental acoustical fan anomaly detection cases. The results demonstrate the superiority of CDDG relative to eight SOTA methods. Furthermore, all comparative methods are integrated into a unified framework, and the first open-source DGFD benchmark is released to promote research in fault diagnosis.

CRediT authorship contribution statement

Linshan Jia: Conceptualization, Methodology, Writing – original draft. **Tommy W.S. Chow:** Writing – review & editing, Supervision, Funding acquisition. **Yixuan Yuan:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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